LIST OF EXPERIMENTS

- An experiment is a scientific procedure that tests a hypothesis or a prediction.
- An experiment usually involves manipulating one or more variables and measuring their effects on other variables.
- An experiment can be classified into different types based on the level of control, the design, the purpose, and the subject matter.
- Some common types of experiments are:
  - Controlled experiment: An experiment in which only one variable is changed at a time, while all other variables are kept constant. This allows the experimenter to isolate the causal effect of the manipulated variable on the outcome variable. For example, testing the effect of different fertilizers on plant growth by applying them to identical plants in identical conditions.
  - Randomized experiment: An experiment in which the subjects or units are randomly assigned to different treatment groups. This ensures that the groups are comparable in all aspects except for the treatment, and reduces the bias and confounding factors that may affect the results. For example, testing the effect of a new drug on patients by randomly assigning them to either receive the drug or a placebo.
  - Natural experiment: An experiment in which the experimenter does not manipulate any variable, but observes the effects of a natural or external event that creates variation in the variables of interest. This allows the experimenter to exploit the natural variation as a source of causal inference, but may also introduce other sources of error and uncertainty. For example, studying the impact of a natural disaster on economic outcomes by comparing the affected and unaffected regions.
  - Quasi-experiment: An experiment in which the experimenter does not have full control over the assignment of subjects or units to different treatment groups, but uses some other method to create or approximate the groups. This may compromise the internal validity of the experiment, but may also increase the external validity or generalizability of the results. For example, evaluating the effect of a policy change on social outcomes by comparing the outcomes before and after the change, or between the regions that implemented the change and those that did not.
  - Field experiment: An experiment that is conducted in a natural or real-world setting, rather than in a laboratory or a controlled environment. This may increase the realism and relevance of the experiment, but may also introduce more noise and variability in the data. For example, testing the effect of a marketing strategy on consumer be-

havior by implementing it in a real store or online platform.

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Note: A minimum of ten experiments from the following should be performed.

- Experiment 1: To study the characteristics of a common emitter transistor amplifier.
- Experiment 2: To study the frequency response of a RC coupled amplifier.
- Experiment 3: To design and implement a Hartley oscillator using transistor.
- Experiment 4: To design and implement a Colpitts oscillator using transistor.
- Experiment 5: To design and implement a phase shift oscillator using op-amp.
- Experiment 6: To design and implement a Wein bridge oscillator using op-amp.
- Experiment 7: To design and implement a voltage regulator using zener diode.
- Experiment 8: To design and implement a half wave and full wave rectifier using diodes.
- Experiment 9: To design and implement a clipper and clamper circuit using diodes.
- Experiment 10: To design and implement a astable and monostable multivibrator using 555 timer.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of hardware based experiments. Here is some content in markdown format that you can use as study material.

(A) Hardware based experiments

Hardware based experiments are experiments that involve the use of physical devices, components, or systems to test a hypothesis, measure a phenomenon, or demonstrate a concept. Hardware based experiments can be classified into different types, such as:

- **Simulation experiments**: These are experiments that use software or hardware models to mimic the behavior of a real system or environment. Simulation experiments can be useful for testing scenarios that are difficult, expensive, or dangerous to perform in reality, such as natural disasters, space exploration, or nuclear reactions. Simulation experiments can also be used to study the effects of changing parameters or variables on the system or environment, such as temperature, pressure, or speed. Simulation experiments can be performed using software tools, such as MATLAB, Simulink, or LabVIEW, or hardware platforms, such as Arduino, Raspberry Pi, or FPGA.

- **Measurement experiments**: These are experiments that use sensors, instruments, or devices to collect data or information about a physical quantity, property, or state. Measurement experiments can be used to verify a theory, validate a model, or calibrate a device. Measurement experiments can also be used to explore the characteristics, behavior, or performance of a system or component, such as voltage, current, resistance, frequency, or power. Measurement experiments can be performed using devices, such as multimeters, oscilloscopes, function generators, or power supplies, or instruments, such as thermometers, barometers, hygrometers, or spectrometers.
- **Design experiments**: These are experiments that involve the creation, modification, or improvement of a hardware system, component, or device. Design experiments can be used to solve a problem, meet a specification, or achieve a goal. Design experiments can also be used to demonstrate the functionality, feasibility, or reliability of a system or device, such as a robot, a circuit, or a sensor. Design experiments can be performed using tools, such as soldering irons, breadboards, or prototyping boards, or components, such as resistors, capacitors, transistors, or LEDs.

1. Verification of Kirchhoff's laws

Kirchhoff's laws are two rules that govern the conservation of charge and energy in electrical circuits. They are named after Gustav Kirchhoff, a German physicist who formulated them in 1845. The two laws are:

- Kirchhoff's current law (KCL): This law states that the algebraic sum of all currents entering and exiting a node must equal zero. A node is a point where two or more branches of a circuit meet. This law implies that charge is conserved at any node, meaning that the current that flows into a node is equal to the current that flows out of it.

- Kirchhoff's voltage law (KVL): This law states that the algebraic sum of all the voltages around any closed loop in a circuit must equal zero. A loop is a path that starts and ends at the same node. This law implies that energy is conserved in any loop, meaning that the total work done by the sources and the loads in a loop is zero.

To verify Kirchhoff's laws experimentally, we need to set up a circuit with a known configuration of resistors, a voltage source, and an ammeter and a voltmeter to measure the currents and voltages in the circuit. We can then apply KCL to any node and KVL to any loop and compare the measured values with the theoretical values calculated using Ohm's law and the resistor combinations rules. The measured values should agree with the theoretical values within the margin of error of the instruments.

A possible circuit diagram for verifying Kirchhoff's laws is shown below:

Circuit diagram

In this circuit, we have four resistors R1, R2, R3, and R4 connected in series and parallel combinations, a voltage source V, and an ammeter A and a voltmeter Vm to measure the currents and voltages. We can label the nodes and the loops as shown in the diagram.

To verify KCL, we can choose any node and sum up the currents entering and exiting that node. For example, at node A, we have:

I1 + I2 - I = 0

where I1 is the current through R1, I2 is the current through R2, and I is the current measured by the ammeter. We can measure I using the ammeter and calculate I1 and I2 using Ohm's law:

I1 = V/R1

I2 = V/R2

where V is the voltage measured by the voltmeter across the voltage source. The measured value of I should be equal to the calculated value of I1 + I2 within the margin of error of the ammeter.

To verify KVL, we can choose any loop and sum up the voltages around that loop. For example, in loop ABCDA, we have:

V - IR1 - IR2 - IR3 = 0

where V is the voltage measured by the voltmeter across the voltage source, I is the current measured by the ammeter, and R1, R2, and R3 are the resistances of the resistors. We can measure V and I using the voltmeter and the ammeter and calculate the voltage drops across the resistors using Ohm's law:

IR1 = I * R1

IR2 = I * R2

IR3 = I * R3

The measured value of V should be equal to the calculated value of IR1 + IR2 + IR3 within the margin of error of the voltmeter.

We can repeat the same procedure for other nodes and loops in the circuit and verify that KCL and KVL are satisfied in each case. This way, we can experimentally verify Kirchhoff's laws for any given circuit.

2. Measurement of power and power factor in a single phase ac series inductive circuit and study improvement of power factor using capacitor

- Power factor is a measure of energy efficiency in an alternating current circuit. It is the ratio of real power (W) to apparent power (VA) or the cosine of the phase angle between voltage and current .
- Real power is the power that is actually consumed by the load and does useful work. Apparent power is the product of the rms values of voltage and current. It is the power that is supplied by the source. Phase angle is the angle by which the current lags or leads the voltage in an ac circuit.
- Power factor can range from 0 to 1. A power factor of 1 means that the voltage and current are in phase and the circuit is purely resistive. A power factor of 0 means that the voltage and current are out of phase by 90 degrees and the circuit is purely inductive or capacitive. A power factor between 0 and 1 means that the circuit has some resistance, inductance and/or capacitance.
- Power factor can be calculated by using the power triangle, which is a right-angled triangle with the sides representing the real power (W), the reactive power (VAR) and the apparent power (VA). The angle of the triangle is the phase angle between voltage and current. The power factor is the cosine of this angle or the ratio of the adjacent side (W) to the hypotenuse (VA).
- Power factor can also be calculated by using the formula: power factor = real power / apparent power = VI cos / VI = cos , where V and I are the rms values of voltage and current and is the phase angle .
- Power factor can be measured by using a single phase power factor meter, which is an instrument that measures the phase angle between voltage and current and displays the power factor on a scale. It consists of two coils, one connected in series with the load and the other connected across the supply. The coils produce torques that are proportional to the current and the voltage respectively. The coils are mounted on a common shaft that rotates according to the net torque. The shaft carries a pointer that indicates the power factor on a calibrated scale.
- Power factor can be improved by using a capacitor in parallel with the load. A capacitor is a device that stores electric charge and opposes the change in voltage. It produces a current that leads the voltage by 90 degrees. When a capacitor is connected in parallel with an inductive load, it reduces the phase angle between voltage and current and increases the power factor.
- The value of the capacitor that is required to improve the power factor can be calculated by using the formula: C = Q / V^2 , where Q is the reactive power that needs to be compensated, V is the rms value of the supply voltage and is the angular frequency of the supply.
- The power and power factor in a single phase ac series inductive circuit can be measured by using a wattmeter, an ammeter and a voltmeter. A wattmeter is an instrument that measures the real power in a circuit. It consists of two coils, one connected in series with the load and the other connected across the supply. The coils produce torques that are proportional to the current and the voltage respectively. The coils are

mounted on a common shaft that rotates according to the net torque. The shaft carries a pointer that indicates the real power on a calibrated scale.

- The ammeter is an instrument that measures the current in a circuit. It is connected in series with the load. The voltmeter is an instrument that measures the voltage across a circuit. It is connected in parallel with the load. The power factor can be calculated by using the formula: power factor = real power / apparent power = wattmeter reading / (ammeter reading x voltmeter reading).
- The improvement of power factor using a capacitor can be studied by connecting a capacitor in parallel with the load and observing the changes in the readings of the wattmeter, the ammeter and the voltmeter. The capacitor should have a suitable value to achieve the desired power factor. The power factor can be calculated by using the same formula as before.

3. Study of phenomenon of resonance in RLC series circuit and obtain resonant frequency.

- A RLC series circuit consists of a resistor (R), an inductor (L) and a capacitor (C) connected in series to an alternating voltage source.
- The current (I) in the circuit is the same for all the components, but the voltage (V) across each component is different and depends on the frequency (f) of the source.
- The voltage across the resistor is in phase with the current and is given by V_R = IR, where I is the rms value of the current.
- The voltage across the inductor leads the current by 90 degrees and is given by V_L = IXL, where XL = 2 fL is the inductive reactance.
- The voltage across the capacitor lags the current by 90 degrees and is given by V_C = IXC, where XC = 1/(2 fC) is the capacitive reactance.
- The total voltage across the circuit is the phasor sum of the individual voltages and is given by V = sqrt((V_R)^2 + (V_L - V_C)^2).
- The impedance (Z) of the circuit is the ratio of the total voltage to the current and is given by Z = V/I = sqrt(R^2 + (XL - XC)^2).
- The phase difference ( ) between the total voltage and the current is given by tan( ) = (V_L - V_C)/V_R.
- The power (P) dissipated in the circuit is given by P = I^2R = VI cos( ).
- Resonance is a phenomenon that occurs when the frequency of the source is such that the inductive reactance and the capacitive reactance are equal, i.e., XL = XC.
- At resonance, the total voltage is equal to the voltage across the resistor, i.e., V = V_R, and the phase difference is zero, i.e., = 0.
- The impedance of the circuit is minimum at resonance and is equal to the resistance, i.e., Z = R.
- The current in the circuit is maximum at resonance and is given by I = V/R, where V is the rms value of the source voltage.
- The power dissipated in the circuit is maximum at resonance and is given

by P = I^2R = V^2/R.
- The resonant frequency (f_0) is the frequency at which resonance occurs and is given by f_0 = 1/(2 sqrt(LC)).
- The quality factor (Q) of the circuit is a measure of the sharpness of the resonance and is given by Q = XL/R = XC/R = 1/(R sqrt(C/L)) at resonance.
- The bandwidth (B) of the circuit is the range of frequencies for which the power dissipated is at least half of the maximum power at resonance and is given by B = f_2 - f_1, where f_2 and f_1 are the frequencies at which P = P_max/2.
- The bandwidth is inversely proportional to the quality factor, i.e., B = f_0/Q.

Hello, I am Sydney, your AI assistant. I can help you with various tasks and topics. You have asked me to write on the topic of connection and measurement of power consumption of a fluorescent lamp (tube light). Here is the content I have generated in markdown format:

4. Connection and measurement of power consumption of a fluorescent lamp (tube light).

A fluorescent lamp (or tube light) is a type of electric light that uses low-pressure mercury vapor and phosphor coating to produce visible light. The mercury vapor emits ultraviolet (UV) radiation, which is converted into visible light by the phosphor coating on the inner surface of the glass tube. The fluorescent lamp requires a ballast to provide the necessary voltage and current to start and maintain the lamp.

The connection and measurement of power consumption of a fluorescent lamp can be done as follows:

- The fluorescent lamp is connected in series with a switch, a fuse, and a wattmeter to the mains supply. The wattmeter measures the power consumed by the lamp in watts (W).
- The ballast is connected in parallel with the lamp. The ballast regulates the current and voltage in the lamp circuit and provides the necessary high voltage to start the lamp. The ballast can be either magnetic or electronic. A magnetic ballast consists of a coil of wire and a capacitor, while an electronic ballast uses a circuit of transistors and diodes.
- The lamp also has a starter, which is a small device that creates a short circuit across the lamp terminals when the switch is turned on. The starter heats up a bimetallic strip, which bends and breaks the short circuit. This causes a high voltage pulse to be generated by the ballast, which ionizes the mercury vapor and initiates the lamp. The starter then cools down and closes the circuit again, allowing the lamp to operate normally. The starter can be either integrated with the lamp or separate from it.
- The power consumption of the fluorescent lamp depends on several factors,

such as the lamp wattage, the ballast type and efficiency, the supply voltage and frequency, and the ambient temperature. The power consumption can be calculated by multiplying the voltage and current measured by the wattmeter. The power factor of the lamp circuit can also be determined by dividing the power consumption by the product of the voltage and current. The power factor indicates how efficiently the lamp circuit uses the power supplied by the mains. A power factor close to 1 means that the lamp circuit is mostly resistive, while a power factor close to 0 means that the lamp circuit is mostly reactive. A low power factor can cause losses and distortions in the power system. Therefore, it is desirable to have a high power factor for the fluorescent lamp circuit. This can be achieved by using an electronic ballast, which has a higher efficiency and a better power factor than a magnetic ballast.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is the content in markdown format:

5. Measurement of power in 3- phase circuit by two-wattmeter method and determination of its power factor for star as well as delta connected load.

- The two-wattmeter method is a technique for measuring the total power in a balanced or unbalanced three-phase circuit using two wattmeters.
- The two wattmeters are connected across two of the three phase voltages, and their readings are added to obtain the total power.
- The power factor of the circuit can be determined by the phase angle between the two wattmeter readings, or by the ratio of the difference and sum of the readings.
- For a star-connected load, the two wattmeters are connected as shown below:

star

- The readings of the two wattmeters are given by:
  - W1 = V1I1 cos(1 - 1) = VphIph cos(1 - 1)
  - W2 = V2I2 cos(2 - 2) = VphIph cos(2 - 2)
- Where Vph and Iph are the phase voltage and current, 1 and 2 are the phase angles of the voltages, and 1 and 2 are the phase angles of the currents.
- The total power is given by:
  - W = W1 + W2 = VphIph [cos(1 - 1) + cos(2 - 2)]
- The power factor is given by:
  - PF = cos(-) = (W1 + W2) / (sqrt(3) VL IL)

- Where and are the average phase angles of the voltages and currents, VL and IL are the line voltage and current.

- For a delta-connected load, the two wattmeters are connected as shown below:

delta

- The readings of the two wattmeters are given by:
  - W1 = V1I1 cos(1 - 1) = VLIL cos(30° + 1 - 1)
  - W2 = V2I2 cos(2 - 2) = VLIL cos(30° + 2 - 2)
- Where VL and IL are the line voltage and current, 1 and 2 are the phase angles of the voltages, and 1 and 2 are the phase angles of the currents.
- The total power is given by:
  - W = W1 + W2 = VLIL [cos(30° + 1 - 1) + cos(30° + 2 - 2)]
- The power factor is given by:
  - PF = cos(-) = (W1 + W2) / (3 VL IL)
  - Where and are the average phase angles of the voltages and currents.

Hello, I am Sydney, your AI assistant. I can help you with your studies. Here is the content on the topic of determination of parameters of ac single phase series RLC circuit.

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##### 6. Determination of parameters of ac single phase series RLC circuit

- An ac single phase series RLC circuit consists of a resistor (R), an inductor (L), and a c
- The parameters of the circuit are the resistance (R), the inductance (L), the capacitance
- The following formulas can be used to determine the parameters of the circuit:

  - The current (I) is the same for all the components in series and is given by Ohm's law:
  - The impedance (Z) is the opposition to the current flow and is given by the Pythagorean
  - The inductive reactance (XL) is the opposition to the current flow due to the inductor a
  - The capacitive reactance (XC) is the opposition to the current flow due to the capacitor
  - The phase angle ( ) is the angle between the voltage and the current and is given by the
  - The power factor (pf) is the cosine of the phase angle and is a measure of how efficient
  - The active power (P) is the power that is dissipated as heat in the resistor and is give
  - The reactive power (Q) is the power that is stored and released by the inductor and the
  - The apparent power (S) is the power that is supplied by the source and is given by: `S =

- To determine the parameters of the circuit, the following steps can be followed:

  1. Measure or obtain the values of R, L, C, and V.
  2. Calculate the frequency (f) of the ac source using the formula: `f = 1/T`, where T is t
  3. Calculate the inductive reactance (XL) and the capacitive reactance (XC) using the form
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4. Calculate the impedance (Z) using the formula: `Z = √(R^2 + (XL - XC)^2)`.
5. Calculate the current (I) using the formula: `I = V/Z`.
6. Calculate the phase angle ( ) using the formula: `  = tan^-1((XL - XC)/R)`.
7. Calculate the power factor (pf) using the formula: `pf = cos( )`.
8. Calculate the active power (P) using the formula: `P = I^2R = VIpf`.
9. Calculate the reactive power (Q) using the formula: `Q = I^2(XL - XC) = VI sin( )`.
10. Calculate the apparent power (S) using the formula: `S = I^2Z = VI`.
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Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for the topic you requested:

7. Determination of (i) Voltage ratio (ii) polarity and (iii) efficiency by load test of a single phase Transformer

- A single phase transformer is a device that transfers electrical energy from one circuit to another through mutual induction.
- The voltage ratio of a transformer is the ratio of the secondary voltage to the primary voltage. It is also equal to the ratio of the number of turns in the secondary winding to the number of turns in the primary winding.
- The polarity of a transformer is the relative direction of the induced voltages in the primary and secondary windings. It can be determined by the dot convention, which assigns a dot to one terminal of each winding. The dots indicate that the voltages at those terminals have the same polarity at any instant.
- The efficiency of a transformer is the ratio of the output power to the input power. It is also equal to the ratio of the output voltage times the output current to the input voltage times the input current, minus the losses in the transformer.
- A load test of a transformer is a method of measuring the voltage ratio, polarity and efficiency of a transformer under different load conditions. The load test consists of connecting a variable load resistor to the secondary winding of the transformer, and measuring the input and output voltages and currents with a voltmeter and an ammeter. The load resistor is varied to obtain different values of output power and power factor.
- The steps for performing a load test of a single phase transformer are:
  1. Connect the primary winding of the transformer to a single phase AC supply of rated voltage and frequency, and the secondary winding to a variable load resistor.
  2. Measure the no-load input voltage V1 and current I0 with a voltmeter and an ammeter, and calculate the no-load input power P0 = V1 x I0.
  3. Adjust the load resistor to obtain a rated output voltage V2 and measure the output current I2 with an ammeter. Calculate the output

power P2 = V2 x I2 and the power factor cos 2 = P2 / (V2 x I2).

4. Measure the input voltage V1 and current I1 with a voltmeter and an ammeter, and calculate the input power P1 = V1 x I1 and the power factor cos 1 = P1 / (V1 x I1).
5. Calculate the voltage ratio K = V2 / V1 and the polarity of the transformer by comparing the dot terminals of the primary and secondary windings.
6. Calculate the efficiency = P2 / P1 and the losses in the transformer Ploss = P1 - P2.
7. Repeat steps 3 to 6 for different values of load resistor and output power, and plot the efficiency versus output power curve for the transformer.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for the topic of determination of efficiency of a dc shunt motor by load test.

8. Determination of efficiency of a dc shunt motor by load test

- A dc shunt motor is a type of dc motor where the field winding is connected in parallel with the armature winding. This means that the field current is independent of the armature current and the speed is relatively constant.
- The efficiency of a dc shunt motor is the ratio of the output power to the input power. The output power is the mechanical power delivered by the motor to the load, and the input power is the electrical power supplied to the motor from the source.
- To determine the efficiency of a dc shunt motor by load test, the following steps are followed:
  - Connect the dc shunt motor to a suitable dc source and a variable load, such as a brake drum or a dynamometer. Also connect an ammeter, a voltmeter, and a wattmeter to measure the armature current, the terminal voltage, and the input power respectively. Connect another ammeter to measure the field current.
  - Start the motor and adjust the load to obtain the desired speed. Note down the readings of the ammeter, the voltmeter, and the wattmeter for the armature circuit, and the ammeter for the field circuit.
  - Calculate the input power as the product of the terminal voltage and the total current (armature current plus field current).
  - Calculate the output power as the product of the torque and the angular speed. The torque can be measured by the load device, such as the brake drum or the dynamometer. The angular speed can be measured by a tachometer or a speedometer.
  - Calculate the efficiency as the ratio of the output power to the input power, expressed as a percentage.

  - Repeat the above steps for different values of load and speed, and plot a graph of efficiency versus output power. The maximum efficiency occurs at the point where the slope of the curve is zero, or where the losses are minimum.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for the topic you requested:

9. To study running and speed reversal of a three phase induction motor and record speed in both directions.

- A three phase induction motor is a type of electric motor that converts alternating current (AC) power into mechanical power by creating a rotating magnetic field in the stator and inducing currents in the rotor.
- The speed of a three phase induction motor depends on the frequency of the AC supply, the number of poles in the stator, and the slip between the rotor and the stator. The speed can be expressed by the formula:

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N = (120f / P) (1 - s)
```

where N is the speed in revolutions per minute (rpm), f is the frequency in hertz (Hz), P is the number of poles, and s is the slip.

- The slip is the difference between the synchronous speed (the speed of the rotating magnetic field) and the actual speed of the rotor. The slip can be expressed as a percentage or a fraction of the synchronous speed. The slip is positive when the rotor is lagging behind the stator, and negative when the rotor is leading ahead of the stator.

- The direction of rotation of a three phase induction motor is determined by the phase sequence of the AC supply. If the phase sequence is ABC, the motor rotates in the clockwise direction. If the phase sequence is ACB, the motor rotates in the anti-clockwise direction. The phase sequence can be changed by interchanging any two wires of the AC supply.

- To study the running and speed reversal of a three phase induction motor, the following steps can be followed:

  - Connect the three phase induction motor to a three phase AC supply through a star-delta starter and a tachometer. The star-delta starter is used to reduce the starting current and the tachometer is used to measure the speed of the motor.
  - Switch on the AC supply and observe the direction and speed of the motor. Note down the readings of the tachometer and the voltmeter. Calculate the slip and the percentage slip of the motor.
  - Switch off the AC supply and interchange any two wires of the AC supply. Switch on the AC supply again and observe the direction and speed of the motor. Note down the readings of the tachometer

  and the voltmeter. Calculate the slip and the percentage slip of the motor.
  - Compare the results and verify that the direction of rotation of the motor is reversed and the speed is slightly changed due to the change in the slip.

10. Demonstration of cut-out sections of machines: dc machine, three phase induction machine, single-phase induction machine and synchronous machine.

- Cut-out sections of machines are parts of machines that are cut or removed to show the internal structure and components of the machines.
- Cut-out sections of machines can be used for demonstration, education, inspection, and maintenance purposes .
- The following are some examples of cut-out sections of machines:
  - DC machine: A DC machine is a machine that converts electrical energy into mechanical energy or vice versa using direct current. A cut-out section of a DC machine can show the commutator-brush arrangement, which is a device that reverses the direction of current in the armature windings to produce a continuous torque. The commutator consists of a set of copper segments that are insulated from each other and connected to the armature coils. The brushes are carbon or graphite blocks that slide on the commutator and make electrical contact with the segments.
  - Three phase induction machine: A three phase induction machine is a machine that converts electrical energy into mechanical energy using alternating current. A cut-out section of a three phase induction machine can show the squirrel cage rotor, which is a type of rotor that consists of a cylindrical laminated core with bars of copper or aluminum embedded in slots and short-circuited at both ends by end rings. The squirrel cage rotor rotates when a rotating magnetic field is induced in the stator windings.
  - Single-phase induction machine: A single-phase induction machine is a machine that converts electrical energy into mechanical energy using single-phase alternating current. A cut-out section of a single-phase induction machine can show the shaded pole or split phase arrangement, which are methods of creating a rotating magnetic field in the stator windings using a single-phase supply. The shaded pole arrangement uses a copper ring or band around a portion of each pole to delay the magnetic flux and create a phase difference. The split phase arrangement uses two sets of stator windings with different resistances and inductances to create a phase difference.
  - Synchronous machine: A synchronous machine is a machine that converts electrical energy into mechanical energy or vice versa using

alternating current at a constant frequency. A cut-out section of a synchronous machine can show the field winding and the armature winding, which are the two main components of the machine. The field winding is the winding that produces the magnetic field in the machine and is usually located on the rotor. The armature winding is the winding that carries the output or input current and is usually located on the stator.

- The following are some diagrams of cut-out sections of machines:
  - DC machine:

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# Experiments available on virtual lab

Virtual lab is a platform that allows users to perform experiments and simulations online, w

- It reduces the cost and time of conducting experiments.
- It enhances the accessibility and availability of experiments for learners and educators.
- It provides a safe and controlled environment for conducting experiments.
- It enables the replication and modification of experiments for different scenarios and par
- It supports the development of analytical and problem-solving skills.

There are many experiments available on virtual lab, covering various disciplines and domain

- Physics: Users can perform experiments on topics such as optics, mechanics, thermodynamics
- Chemistry: Users can perform experiments on topics such as organic chemistry, inorganic ch
- Biology: Users can perform experiments on topics such as cell biology, molecular biology,
- Engineering: Users can perform experiments on topics such as civil engineering, mechanical



##### 1. Kirchhoff s laws.

Kirchhoff's laws are a set of two laws that describe the conservation of charge and energy i

- Kirchhoff's current law (KCL): This law states that the algebraic sum of currents entering
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- Kirchhoff's voltage law (KVL): This law states that the algebraic sum of voltages around a

These two laws are the foundation of circuit analysis, as they allow us to find the unknown

Here are some diagrams to illustrate Kirchhoff's laws:

KCL diagram

This diagram shows Kirchhoff's current law applied to a node with four branches. The current

I1 + I2 + I3 + I4 = 0

This equation can be used to find any unknown current in the node.

KVL diagram

This diagram shows Kirchhoff's voltage law applied to a loop with four components. The volta

V1 + V2 + V3 + V4 = 0

This equation can be used to find any unknown voltage in the loop.

Hello, I am Sydney, your AI assistant. I can help you with your query. Here is some informat

Virtual Lab

- Virtual Lab is a project initiated by the Ministry of Education, Government of India, unde
- The project aims to provide remote access to laboratories in various disciplines of scienc
- The project is a collaborative effort of 12 participating institutes, including Amrita Vis
- Amrita Vishwa Vidyapeetham has developed over 220 simulation-based experiments for the Vir
- The Virtual Lab experiments are designed to enhance the learning outcomes of students by p
- The Virtual Lab experiments are accessible through a web portal (http://vlab.amrita.edu/)
- The Virtual Lab project also provides training and support to teachers and students throug
- The Virtual Lab project has received several awards and recognitions, such as the World Ed

Virtual Lab Experiment: Study of Osmosis

- Osmosis is the process of movement of water molecules across a selectively permeable membr
- The water potential of a solution is determined by its solute concentration and pressure.
- Osmosis is important for the survival of cells, as it helps to maintain the water balance
- In this experiment, you will study the effect of osmosis on plant cells (potato tuber cell
- You will observe the changes in the mass, length, and appearance of the potato tuber cells
- You will also calculate the percentage change in mass and length of the potato tuber cells

- You will use the following materials and equipment for this experiment:

  - Potato tuber
  - Cork borer
  - Knife
  - Electronic balance
  - Ruler
  - Beakers
  - Distilled water
  - Sucrose solutions of different concentrations (0.2 M, 0.4 M, 0.6 M, 0.8 M, and 1.0 M)
  - Test tubes
  - Test tube rack
  - Pipette
  - Sheep blood
  - Compound microscope
  - Glass slides
  - Cover slips
  - Forceps
  - Dropping bottle
  - Safranin stain
  - Filter paper
  - Stopwatch
  - Calculator

- You will follow the following steps for this experiment:

  - Cut a potato tuber into small pieces using a knife.
  - Use a cork borer to cut out six cylindrical pieces of potato tuber of equal diameter.
  - Cut the potato cylinders to a length of 3 cm using a knife and a ruler.
  - Weigh each potato cylinder using an electronic balance and record its mass in a table.
  - Label six beakers as A, B, C, D, E, and F.
  - Fill beaker A with 100 ml of distilled water and beakers B, C, D, E, and F with 100 ml o
  - Place one potato cylinder in each beaker and start the stopwatch.
  - After 30 minutes, remove the potato cylinders from the beakers and blot them dry with fi
  - Weigh each potato cylinder again using an electronic balance and record its mass in the
  - Measure the length of each potato cylinder using a ruler and record it in the same table
  - Calculate the percentage change

Hello, I am Sydney, your AI assistant. I can help you with your query.

Thevenin's theorem is a useful method to simplify any linear circuit, regardless of its comp

The steps to apply Thevenin's theorem are:

1. Identify the terminals A and B of the circuit where the load is connected.
2. Remove the load resistor and replace it with an open circuit.
3. Calculate the Thevenin voltage, Vth, which is the voltage across the open circuit termina
4. Calculate the Thevenin resistance, Rth, which is the equivalent resistance seen from the
5. Connect the Thevenin voltage and the Thevenin resistance in series to form the Thevenin e
6. Reconnect the load resistor to the terminals A and B of the Thevenin equivalent circuit a

The advantages of using Thevenin's theorem are:

- It reduces the complexity of the circuit and makes the analysis easier.
- It allows us to find the current, voltage, and power across any load resistor without char
- It can be applied to both DC and AC circuits.

The limitations of using Thevenin's theorem are:

- It is only applicable to linear circuits, which means the circuit elements must obey Ohm's
- It is not applicable to circuits with dependent sources, unless the dependent sources are
- It may not be practical to find the Thevenin resistance for some circuits with many resist

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to discuss. Yo

Virtual Lab: Study of Osmosis

Osmosis is the process of movement of water molecules across a selectively permeable membran

In this virtual lab, you will learn how to:

- Define osmosis and water potential
- Explain the factors that affect the rate of osmosis
- Perform an experiment to demonstrate osmosis using potato tubers
- Analyze the data and draw conclusions

Theory

Water potential is the measure of the tendency of water molecules to move from one place to

Solute potential is the effect of dissolved solutes on the water potential. It is always neg

Pressure potential is the effect of physical pressure on the water potential. It can be posi

The water potential of a system is equal to the sum of the solute potential and the pressure

Water potential = Solute potential + Pressure potential

Osmosis occurs when there is a difference in water potential between two solutions separated

The rate of osmosis depends on several factors, such as:

- The difference in water potential between the two solutions
- The surface area of the membrane
- The permeability of the membrane
- The temperature of the system

Experiment

In this experiment, you will use potato tubers as osmometers to demonstrate osmosis. Potato

You will prepare different concentrations of sucrose solution and immerse potato tubers in t

The steps of the experiment are as follows:

1. Prepare six different concentrations of sucrose solution: 0.0 M (distilled water), 0.1 M,
2. Peel six potato tubers and cut them into cylindrical pieces of equal size. Use a ruler an
3. Measure the initial mass and length of each potato tuber using a balance and a ruler. Rec
4. Immerse each potato tuber in a beaker containing 100 mL of the corresponding sucrose solu
5. After 24 hours, remove the potato tubers from the sucrose solutions and blot them dry wit
6. Calculate the percentage change in mass and length of each potato tuber using the followi

Percentage change = (Final value - Initial value) / Initial value x 100

Record the data in a table.
7. Plot a graph of percentage change in mass and length versus sucrose concentration. Use a
8. Determine the

3. RLC series resonance

- An RLC series circuit consists of a resistor (R), an inductor (L), and a capacitor (C) con
- The current in the circuit is the same for all the components, but the voltage across each
- The impedance of the circuit is given by Z = R + j(XL - XC), where XL = 2 fL is the induct
- The phase angle of the circuit is given by = tan^-1((XL - XC)/R), where is positive if
- The power factor of the circuit is given by cos , which is the ratio of the real power to
- Series resonance occurs when the inductive reactance is equal to the capacitive reactance,
- At resonance, the impedance of the circuit is minimum and equal to R, the phase angle is z
- At resonance, the current in the circuit is maximum and equal to V/R, where V is the sourc
- At resonance, the voltage across the inductor and the capacitor are equal and opposite in
- A series resonant circuit can draw heavy current and power from the source, and it can act

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or di

- Virtual lab is an initiative of the Ministry of Education, Government of India, under the
- Virtual lab aims to provide remote access to labs in various disciplines of science and en
- Virtual lab also provides a complete learning management system, where you can find online
- Virtual lab covers various domains such as biotechnology, chemical engineering, civil engi
- Virtual lab consists of two components: the web portal and the lab server. The web portal
- Virtual lab is developed and maintained by a consortium of 12 institutions, led by Amrita
- Virtual lab is free and open to all, and you can access it from any device with an interne
- Virtual lab is a useful resource for enhancing your learning outcomes, developing your ski

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some informat

4. Measurement of power in 3- phase circuit by two wattmeter method and determination

- The two wattmeter method is a technique for measuring the total power in a three-phase cir
- The current coils of the two wattmeters are connected in series with any two line conducto
- The connection diagram of two wattmeter method for a star-connected load is shown below:

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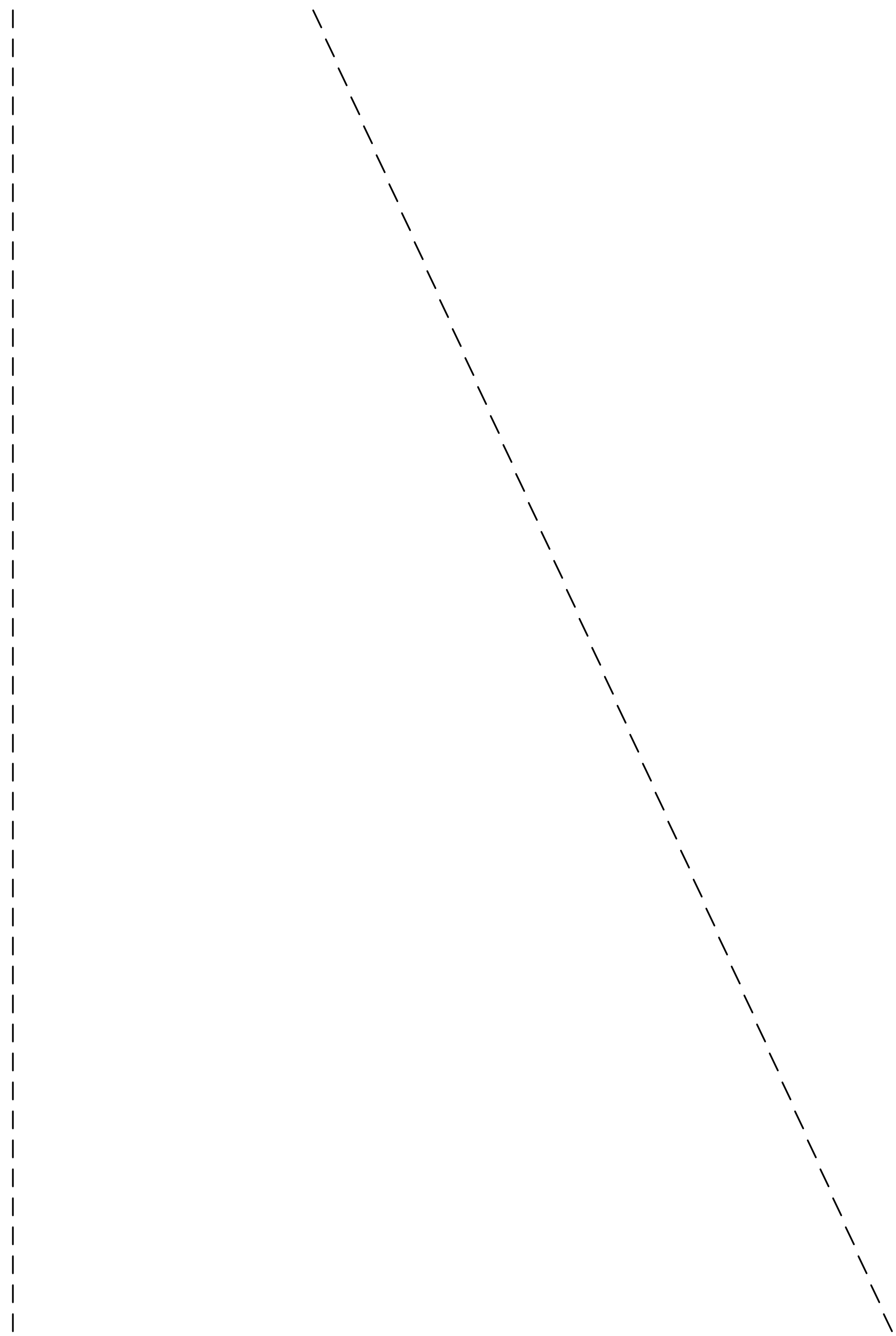

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Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some content in markdown format that you can use for your study material.

Virtual lab: Power measurement using Dreamweaver

- A virtual lab is a computer-based simulation of a real laboratory that allows students to perform experiments and learn concepts without the need for physical equipment, materials, or space .
- Dreamweaver is a web development software that can be used to create interactive web pages and applications.
- Power measurement is the process of determining the amount of electrical energy consumed or delivered by a circuit or a device.
- In this virtual lab, you will learn how to measure power using different devices and techniques, such as:
  - Ammeter: A device that measures the electric current in a circuit. It is connected in series with the circuit element whose current is to be measured.
  - Voltmeter: A device that measures the electric potential difference or voltage across two points in a circuit. It is connected in parallel with the circuit element whose voltage is to be measured.
  - Wattmeter: A device that measures the power in a circuit. It consists of two coils: a current coil and a potential coil. The current coil is connected in series with the load, and the potential coil is connected in parallel with the load.
  - Power factor meter: A device that measures the power factor of a circuit. The power factor is the ratio of the real power to the apparent power in a circuit. It indicates how efficiently the circuit uses the supplied power.
- You will also learn how to use Dreamweaver to create a web page that displays the results of your power measurement experiments. You will use the following features of Dreamweaver:

  – Design view: A graphical interface that allows you to create and edit web pages visually.
  – Code view: A text editor that allows you to write and edit the HTML, CSS, and JavaScript code of your web pages.
  – Split view: A mode that shows both the design view and the code view of your web page simultaneously.
  – Insert panel: A toolbar that provides quick access to various elements and objects that you can insert into your web page, such as images, tables, forms, etc.
  – Properties panel: A panel that shows the properties and attributes of the selected element or object in your web page, such as width, height, color, alignment, etc.
  – Preview in browser: A feature that allows you to test and view your web page in different web browsers, such as Chrome, Firefox, Safari, etc.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for the topic 5. Determination of parameters of ac single phase series RLC circuit.

```
# 5. Determination of parameters of ac single phase series RLC circuit

- An ac single phase series RLC circuit consists of a resistor (R), an inductor (L), and a c
- The current (I) in the circuit is the same for all the components, but the voltage across
- The impedance (Z) of the circuit is the total opposition to the current flow, and it is gi

  Z = R + j(X_L - X_C)

  where j is the imaginary unit, X_L is the inductive reactance, and X_C is the capacitive r

- The inductive reactance (X_L) is the opposition to the current flow due to the self-induct

  X_L = 2 fL

  where f is the frequency of the source, and L is the inductance of the coil.

- The capacitive reactance (X_C) is the opposition to the current flow due to the charge sto

  X_C = 1/(2 fC)

  where C is the capacitance of the capacitor.

- The phase angle ( ) of the circuit is the angle between the voltage and the current, and i

    = tan^(-1)((X_L - X_C)/R)

  The phase angle indicates whether the circuit is inductive (  > 0), capacitive (  < 0), or
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- The power factor (pf) of the circuit is the ratio of the true power (P) to the apparent po

  pf = P/S = cos( )

  The power factor indicates how efficiently the circuit uses the power supplied by the sour

- The parameters of the ac single phase series RLC circuit can be determined by measuring th
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Virtual lab

- Virtual lab is a web-based platform that provides remote access to various experiments in science and engineering disciplines.
- Virtual lab aims to enhance the learning outcomes of students by providing them with a realistic and interactive simulation of a real lab environment.
- Virtual lab also enables students to perform experiments that may not be feasible or safe in a physical lab, such as nuclear physics, biotechnology, or nanotechnology.
- Virtual lab is developed by a consortium of institutions, led by Amrita Vishwa Vidyapeetham, under the National Mission on Education through ICT (NMEICT) of the Ministry of Education, Government of India.
- Virtual lab consists of over 220 simulation-based experiments, covering 12 domains of science and engineering, such as physics, chemistry, biology, computer science, electrical engineering, mechanical engineering, etc.
- Virtual lab also provides animations, videos, graphics, and quizzes to supplement the simulation and enhance the understanding of the concepts and principles involved in the experiments.
- Virtual lab is accessible to anyone with an internet-enabled computer or mobile device, and does not require any installation or registration.
- Virtual lab is designed to be user-friendly, interactive, and self-explanatory, and provides feedback and guidance to the users throughout the experiment.
- Virtual lab is intended to complement the existing physical labs and curriculum, and not to replace them. It also provides an opportunity for collaborative learning and peer interaction among students and teachers.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have asked me to write on the topic of observing the B-H loop of a ferromagnetic material in CRO. Here is the content I have generated for you in markdown format:

```
# 6. To observe the B-H loop of a ferromagnetic material in CRO.

## Aim
To observe the B-H loop or hysteresis loop of a ferromagnetic material in a cathode ray osci

## Theory
```

```
- A ferromagnetic material is a material that can be magnetized by an external magnetic fiel
- The relationship between the magnetic field intensity H and the magnetic flux density B of
- A B-H loop or hysteresis loop is a graphical representation of the cyclic magnetization pr
- A B-H loop can be observed in a CRO by using a solenoid as the primary coil and a toroidal
- The AC source produces a sinusoidal magnetic field in the primary coil, which induces a si
- The CRO displays the voltage of the secondary coil on the vertical axis (Y-axis) and the c
- The CRO trace shows a closed loop that represents the B-H loop of the material. The shape

## Apparatus
- A solenoid with a variable AC source
- A toroidal core of a ferromagnetic material with a secondary coil
- A CRO with probes and leads
- A multimeter

## Procedure
- Connect the solenoid to the AC source and the multimeter. Adjust the AC source to a suitab
- Insert the toroidal core inside the solenoid and connect the secondary coil to the CRO. Se
- Observe the CRO trace and note the shape and size of the B-H loop. Measure the peak-to-pea
- Repeat the steps with different amplitudes and frequencies of the AC source and observe th

## Observations
- The B-H loop is a closed loop that is symmetrical about the origin. It has a clockwise dir
- The B-H loop has a finite width, which indicates the presence of hysteresis in the materia
- The B-H loop has a saturation region, where the B value reaches a maximum and does not cha
- The B-H loop has a remanence point, where the B value is nonzero when the H value is zero.
- The B-H loop has a coercivity point, where the H value is nonzero when the B value is zero
- The B-H loop changes with the amplitude and frequency of the AC source. As the amplitude i

## Conclusion
The B-H loop of a ferromagnetic material can be observed in a CRO by using a solenoid and a
```

Hello, I am Sydney, your AI assistant. I can help you with your query.

The topic you have chosen is **Virtual lab**. Here is some information about it:

- Virtual lab is a web-based platform that provides remote access to various experiments in science and engineering disciplines.
- Virtual lab aims to enhance the learning outcomes of students by providing them with a realistic and interactive simulation of a real lab environment.
- Virtual lab also helps students who do not have access to physical labs or equipment due to geographical or economic constraints.
- Virtual lab is developed by Amrita Vishwa Vidyapeetham, a leading university in India, in collaboration with other institutions and organizations.
- Virtual lab offers over 220 simulation-based experiments in various domains, such as physics, chemistry, biology, biotechnology, computer science, electrical engineering, mechanical engineering, etc.
- Virtual lab also provides animations, graphics, videos, quizzes, and feed-

back to enrich the learning experience of students.
- Virtual lab is accessible through the website https://vlab.amrita.edu/ or the mobile app Amrita Online Lab.

Some of the benefits of using virtual lab are:

- It enables students to perform experiments at their own pace and convenience, without any time or location constraints.
- It reduces the cost and risk of conducting experiments in physical labs, especially for hazardous or expensive materials or equipment.
- It enhances the conceptual understanding and problem-solving skills of students by providing them with multiple scenarios and parameters to manipulate and observe.
- It fosters the development of scientific inquiry and curiosity among students by allowing them to explore and experiment with various phenomena and concepts.
- It supports the teachers and instructors by providing them with a ready-made and standardized resource for teaching and assessment.

Some of the challenges of using virtual lab are:

- It may not be able to replicate the exact feel and touch of a physical lab, especially for some tactile or sensory experiments.
- It may require a reliable internet connection and a compatible device to access and run the simulations smoothly and effectively.
- It may not be able to address all the learning objectives and outcomes of a particular experiment or course, and may need to be supplemented with other methods or materials.
- It may not be able to capture the social and collaborative aspects of learning in a physical lab, such as peer interaction and feedback, group work, etc.

7. Determination of the efficiency of a dc motor by loss summation method (Swinburne's test).

- The efficiency of a dc motor is defined as the ratio of output (mechanical) power to input (electrical) power.

- The output power of a dc motor can be measured by using a dynamometer or a brake, which applies a torque and measures the rotational speed of the motor shaft.

- The input power of a dc motor can be measured by using a power analyzer, which measures the voltage and current supplied to the motor.

- The efficiency of a dc motor can be calculated by using the formula:

  ```
  Efficiency = Output power / Input power
  ```

- However, measuring the output power of a dc motor on load requires a suitable load device, which may not be available or convenient for large motors.
- An alternative method of measuring the efficiency of a dc motor is to determine its losses (instead of measuring the output power on load) and then use the formula:

  `Efficiency = (Input power - Losses) / Input power`
- This method is called the loss summation method or the Swinburne's test.
- The loss summation method enables the determination of losses without actually loading the motor. The power is required to supply the losses only, so there is no difficulty in applying this method even to very large motors.
- The losses in a dc motor can be classified into two types: constant losses and variable losses .
- Constant losses are those losses that do not vary with the load, such as core loss, friction loss and windage loss .
- Variable losses are those losses that vary with the load, such as copper loss in the armature and the field winding .
- The loss summation method involves the following steps :
  - Run the motor at no load and measure the input power, the no-load current and the speed.
  - Calculate the constant losses by multiplying the input power by the efficiency at no load, which can be assumed to be 0.8 for shunt motors and 0.75 for compound motors.
  - Calculate the copper loss in the field winding by multiplying the field resistance and the square of the field current, which can be assumed to be constant for shunt and compound motors.
  - Calculate the copper loss in the armature at no load by multiplying the armature resistance and the square of the no-load current.
  - Calculate the variable loss at any load by multiplying the armature resistance and the square of the load current, which can be obtained from the no-load current and the rated current of the motor.
  - Calculate the total loss at any load by adding the constant loss and the variable loss.
  - Calculate the efficiency at any load by subtracting the total loss from the input power and dividing by the input power.
- The advantages of the loss summation method are:
  - It is simple and convenient to perform, as it does not require a load device or a dynamometer.
  - It is economical, as it consumes less power than the direct method.

  - It is applicable to any size of motor, as it does not depend on the availability of a suitable load.

- The disadvantages of the loss summation method are:
  - It is not very accurate, as it involves some assumptions and approximations, such as the efficiency at no load, the field current and the armature resistance.
  - It does not account for the stray load losses, which are the additional losses due to the distortion of the magnetic field and the eddy currents in the armature at high loads.
  - It does not provide the actual output power and torque of the motor on load, which may be required for some applications.

Course Outcomes:

- Course outcomes are statements that describe what students are expected to know and be able to do by the end of a course.
- Course outcomes are aligned with the course objectives, which are derived from the program outcomes and the institutional mission and vision.
- Course outcomes are measurable, specific, and student-centered. They focus on the knowledge, skills, and attitudes that students will demonstrate as a result of the course.
- Course outcomes are used to guide the design of the course content, activities, assessments, and feedback. They also help to evaluate the effectiveness of the course and the achievement of the students.
- Course outcomes are communicated to the students at the beginning of the course and throughout the course. They help to motivate the students and to monitor their own progress and learning.

Course Outcome (CO) Bloom's Level

- A course outcome (CO) is a statement that describes what students should be able to do or demonstrate by the end of a course.
- A CO should be specific, measurable, achievable, relevant, and time-bound (SMART).
- A CO should also align with the program outcomes (POs) and the course objectives (COs).
- A CO can be written using the Bloom's taxonomy of cognitive domains, which classifies learning outcomes into six levels: knowledge, comprehension, application, analysis, synthesis, and evaluation.
- The Bloom's level of a CO indicates the degree of cognitive complexity and depth of learning that is expected from the students.
- The higher the Bloom's level, the more challenging and sophisticated the CO is.

- The Bloom's level of a CO should match the level of assessment that is used to measure the students' achievement of the CO.
- The Bloom's level of a CO can be determined by using appropriate verbs that reflect the cognitive domain of the CO.
- Some examples of verbs for each Bloom's level are:
  - Knowledge: define, identify, list, name, recall, recognize, state, etc.
  - Comprehension: describe, explain, interpret, paraphrase, summarize, etc.
  - Application: apply, demonstrate, illustrate, perform, solve, use, etc.
  - Analysis: analyze, compare, contrast, differentiate, examine, etc.
  - Synthesis: create, design, develop, generate, integrate, etc.
  - Evaluation: appraise, assess, critique, evaluate, judge, etc.
- An example of a CO with its Bloom's level is:
  - CO: By the end of this course, the student will be able to design and implement a database system for a given problem domain using SQL and ER diagrams.
  - Bloom's level: Synthesis

At the end of this course, the students should be able to:

- Explain the basic concepts and principles of artificial intelligence, such as agents, search, knowledge representation, reasoning, planning, learning, natural language processing, computer vision, and robotics.
- Apply various artificial intelligence techniques to solve problems, such as heuristic search, constraint satisfaction, logic programming, probabilistic inference, machine learning, natural language processing, computer vision, and robotics.
- Evaluate the strengths and limitations of different artificial intelligence methods and tools, and compare their performance and applicability to different domains and scenarios.
- Design and implement simple artificial intelligence systems using programming languages and frameworks, such as Python, Prolog, TensorFlow, PyTorch, OpenCV, and ROS.
- Demonstrate ethical awareness and social responsibility when developing and using artificial intelligence systems, and consider the potential impacts and implications of artificial intelligence on society, economy, environment, and human values.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some content that you can use for your study material.

CO 1 Conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits. K3

- KVL stands for Kirchhoff's Voltage Law, which states that the algebraic sum of the voltages around any closed loop in a circuit is zero. KCL stands for Kirchhoff's Current Law, which states that the algebraic sum of the currents entering and leaving any node in a circuit is zero.
- Network theorems are mathematical tools that can be used to simplify and analyze complex DC circuits. Some of the common network theorems are superposition theorem, Thevenin's theorem, Norton's theorem, maximum power transfer theorem, and reciprocity theorem.
- To conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits, you will need some basic components such as resistors, voltage sources, current sources, ammeters, voltmeters, and breadboards. You will also need a multimeter to measure the voltage and current values in the circuit.
- Here are some examples of experiments that you can perform:
  - Experiment 1: Verify KVL and KCL in a simple series-parallel circuit. Connect three resistors of different values in series and parallel combinations, and connect a voltage source across the circuit. Use an ammeter to measure the current in each branch, and use a voltmeter to measure the voltage across each resistor. Apply KVL and KCL to the circuit and compare the theoretical values with the measured values. You should find that they are equal or very close, within the margin of error of the instruments.
  - Experiment 2: Apply superposition theorem to a circuit with two independent sources. Connect two voltage sources of different values in series with a resistor, and connect another resistor in parallel with the second source. Use a voltmeter to measure the voltage across the parallel resistor. Apply superposition theorem to the circuit by turning off one source at a time and calculating the voltage across the parallel resistor due to each source separately. Then add the two voltages to get the total voltage. Compare the theoretical value with the measured value. You should find that they are equal or very close, within the margin of error of the instruments.
  - Experiment 3: Apply Thevenin's theorem to a circuit with a load resistor. Connect a voltage source in series with two resistors, and connect another resistor as the load across the second resistor. Use a voltmeter to measure the voltage across the load resistor. Apply Thevenin's theorem to the circuit by finding the Thevenin equivalent voltage and resistance of the circuit without the load resistor. Then connect the load resistor to the Thevenin equivalent circuit and calculate the voltage across the load resistor. Compare the theoretical value with the measured value. You should find that they are equal or very close, within the margin of error of the instruments.
  - Experiment 4: Apply Norton's theorem to a circuit with a load resistor. Connect a current source in parallel with two resistors, and

connect another resistor as the load in series with the second resistor. Use an ammeter to measure the current through the load resistor. Apply Norton's theorem to the circuit by finding the Norton equivalent current and resistance of the circuit without the load resistor. Then connect the load resistor to the Norton equivalent circuit and calculate the current through the load resistor. Compare the theoretical value with the measured value. You should find that they are equal or very close, within the margin of error of the instruments.

- Experiment 5: Apply maximum power transfer theorem to a circuit with a variable load resistor. Connect a voltage source in series with a resistor, and connect a variable resistor as the load across the source. Use a voltmeter and an ammeter to measure the voltage and current across the load resistor. Vary the value of the load resistor and record the power dissipated by the load resistor for each value. Apply maximum power transfer theorem to the circuit by finding the value of the load resistor that matches the internal resistance of the source. Then calculate the maximum power that can be transferred to the load resistor. Compare the theoretical value with the measured value. You should find that they are equal or very close, within the margin of error of the instruments.
- Experiment 6: Apply reciprocity theorem to a circuit with two terminals. Connect a voltage source in series with two resistors, and connect another resistor in parallel with the second resistor. Connect two terminals across the parallel resistor. Use a voltmeter to measure the voltage between the terminals. Then swap the positions of the source and the terminals, and measure the

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have asked me to write on the topic of CO 2 Demonstrate the behavior of AC circuits connected to single phase AC supply and measure power in single phase as well as three phase electrical circuits. K4. Here is the content I have written in markdown format:

```
# CO 2 Demonstrate the behavior of AC circuits connected to single phase AC supply and measu

## Introduction
- An AC circuit is a circuit that carries alternating current (AC), which is the type of cur
- A single phase AC supply is a source of AC voltage that has only one phase, which means th
- A three phase AC supply is a source of AC voltage that has three phases, which means that
- The behavior of AC circuits depends on the type and combination of components that are con
- The power in AC circuits is the rate of energy transfer from the AC supply to the componen

## Behavior of AC circuits connected to single phase AC supply
- The behavior of AC circuits connected to single phase AC supply can be analyzed by using t
- A phasor is a vector that represents the magnitude and phase angle of an AC quantity, such
- The impedance of an AC circuit is the ratio of the phasor voltage to the phasor current. T
- The admittance of an AC circuit is the reciprocal of the impedance. The admittance can be
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- The voltage and current in an AC circuit can be expressed in terms of their peak values, r
- The power factor of an AC circuit is the ratio of the real power to the apparent power. Th
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CO 3 Perform experiment illustrating BH curve of magnetic materials. K3

- The BH curve of a magnetic material is a plot of the magnetic flux density B versus the magnetic field intensity H.
- The BH curve shows the relationship between the magnetization of the material and the applied magnetic field.
- The BH curve can be used to determine the magnetic properties of the material, such as permeability, coercivity, saturation, hysteresis, and remanence.
- To perform the experiment, the following steps are required:
  - Prepare a solenoid coil of known number of turns and length, and connect it to a variable DC power supply and an ammeter.
  - Insert a magnetic material (such as iron, steel, or ferrite) inside the coil, and measure its length and cross-sectional area.
  - Connect a search coil of known number of turns and area to an AC voltmeter, and place it around the magnetic material.
  - Vary the current in the solenoid coil, and measure the corresponding voltage in the search coil.
  - Calculate the magnetic field intensity H using the formula $H = NI/L$, where N is the number of turns, I is the current, and L is the length of the solenoid coil.
  - Calculate the magnetic flux density B using the formula $B = V/(NAf)$, where V is the voltage, N is the number of turns, A is the area, and f is the frequency of the search coil.
  - Plot the BH curve by taking B on the y-axis and H on the x-axis.
  - Observe the shape and features of the BH curve, and compare it with the theoretical values and curves for different types of magnetic materials.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some information on the efficiency of a single phase transformer:

- The efficiency of a single phase transformer is defined as the ratio of the output power to the input power, expressed as a percentage. It indicates how well the transformer converts the electrical energy from the primary side to the secondary side.
- The efficiency of a single phase transformer depends on the load current and the power factor of the load. The efficiency is maximum when the load current is such that the copper losses (due to the resistance of the

windings) are equal to the core losses (due to the hysteresis and eddy currents in the magnetic material).

- The efficiency of a single phase transformer can be calculated by the following formula:

  ```
  = (Output power) / (Input power) * 100%
  ```

  where

  ```
  Output power = (Secondary voltage) * (Secondary current) *
  (Power factor)
  ```

  ```
  Input power = (Primary voltage) * (Primary current)
  ```

- The efficiency of a single phase transformer can also be calculated by the following formula:

  ```
  = (Output power) / (Output power + Losses) * 100%
  ```

  where

  ```
  Losses = Copper losses + Core losses
  ```

  ```
  Copper losses = (Primary resistance) * (Primary current)^2 +
  (Secondary resistance) * (Secondary current)^2
  ```

  ```
  Core losses = Hysteresis losses + Eddy current losses
  ```

- The efficiency of a single phase transformer is usually in the range of 95% to 99%. For large power transformers with very low losses, the efficiency can be as high as 99.7%.

CO 5 Perform experiments on speed measurement and reversal of direction of three phase induction motor and Identify the type of DC and AC machines based on their construction. K4

- Speed measurement of three phase induction motor:
  - One method of measuring the speed of a three phase induction motor is to use a tachometer, which is a device that measures the rotational speed of a shaft by counting the number of pulses per unit time generated by a sensor attached to the shaft.
  - Another method of measuring the speed of a three phase induction motor is to use a stroboscope, which is a device that produces flashes of light at a variable frequency. By adjusting the frequency of the flashes to match the rotational speed of the motor, the motor appears to be stationary or moving slowly. The frequency of the flashes can then be read from a scale on the stroboscope.
- Reversal of direction of three phase induction motor:

- The direction of rotation of a three phase induction motor is determined by the phase sequence of the voltage applied to the stator windings. To reverse the direction of rotation, any two of the three stator leads can be interchanged .
- For example, if the stator leads are labeled as L1, L2, and L3, and the motor rotates clockwise when connected as L1-L2-L3, then the motor will rotate counterclockwise when connected as L1-L3-L2, L2-L1-L3, or L3-L2-L1.
- A reverse starter is a device that can switch the stator leads automatically to reverse the direction of rotation of the motor. It consists of two contactors, one for forward rotation and one for reverse rotation, and an interlock circuit that prevents both contactors from being energized at the same time.

- Identification of the type of DC and AC machines based on their construction:
  - DC machines are electric machines that convert electrical energy into mechanical energy (DC motors) or mechanical energy into electrical energy (DC generators) using direct current. DC machines have two main parts: a stator, which is the stationary part that contains the field windings, and a rotor, which is the rotating part that contains the armature windings.
  - AC machines are electric machines that convert electrical energy into mechanical energy (AC motors) or mechanical energy into electrical energy (AC generators) using alternating current. AC machines have two main types: synchronous machines and induction machines.
  - Synchronous machines are AC machines that operate at a constant speed that is proportional to the frequency of the supply voltage. Synchronous machines have two main parts: a stator, which is the stationary part that contains the armature windings, and a rotor, which is the rotating part that contains the field windings or permanent magnets.
  - Induction machines are AC machines that operate at a variable speed that depends on the load and the slip. Induction machines have two main parts: a stator, which is the stationary part that contains the primary windings, and a rotor, which is the rotating part that contains the secondary windings or short-circuited conductors.

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

These are the six levels of cognitive learning according to Bloom's taxonomy, a framework for classifying educational objectives and outcomes. They are arranged from lower-order to higher-order thinking skills, and they represent the degree of complexity and depth of learning expected from students.

- K1 – Remember: This level involves recalling facts, terms, definitions, concepts, or procedures from memory. Examples of verbs used at this level are: define, list, name, identify, label, recognize, etc.
- K2 – Understand: This level involves explaining the meaning, interpretation, or summary of information in one's own words. Examples of verbs used at this level are: describe, explain, paraphrase, summarize, illustrate, classify, compare, etc.
- K3 – Apply: This level involves using learned information to solve problems or perform tasks in new or familiar situations. Examples of verbs used at this level are: apply, use, demonstrate, solve, calculate, implement, etc.
- K4 – Analyze: This level involves breaking down information into parts, examining the relationships, causes, effects, or implications of each part, and identifying patterns or trends. Examples of verbs used at this level are: analyze, compare, contrast, differentiate, distinguish, examine, etc.
- K5 – Evaluate: This level involves making judgments, assessments, or critiques based on criteria, standards, or evidence. Examples of verbs used at this level are: evaluate, judge, critique, appraise, argue, justify, etc.
- K6 – Create: This level involves generating, producing, or designing new or original ideas, products, or solutions based on existing or synthesized information. Examples of verbs used at this level are: create, design, develop, construct, compose, invent, etc.